4. The cover slip is placed on top of the counting chamber.
5. The tip of the pipette is placed on the counting chamber and fluid discharged till it fills the chambers.
6. The cells are allowed to settle for several minutes and the ruled area is viewed under the microscope.

A person suffering from severe fever, head ach and fatigue so he visits his doctor. What type of diagnosing procedure does the doctor recommended for him. Name the medical equipment is used for it?

Preparation of Blood Smear

The examination of peripheral dry blood smear is a very important laboratory test as it is possible to

- Estimate approximately the number of cellular components
- Study the morphology of these components
- Observe the presence of blood parasites
- Study the response of the body to various diseases

The methodology for the preparation of blood smear is as follows (Figure12.11)

1. Place a drop of blood on a clean glass slide about 1cm from one end
2. Using another glass slide placed at an angle of about 45° to the previous slide.
3. Spread the drop of blood quickly in one stroke as a thin film
4. Stain the film using Leishman's stain
5. Allow the slide to dry and wash the excess stain
6. Observe the slide under a light microscope (Figure12.12).

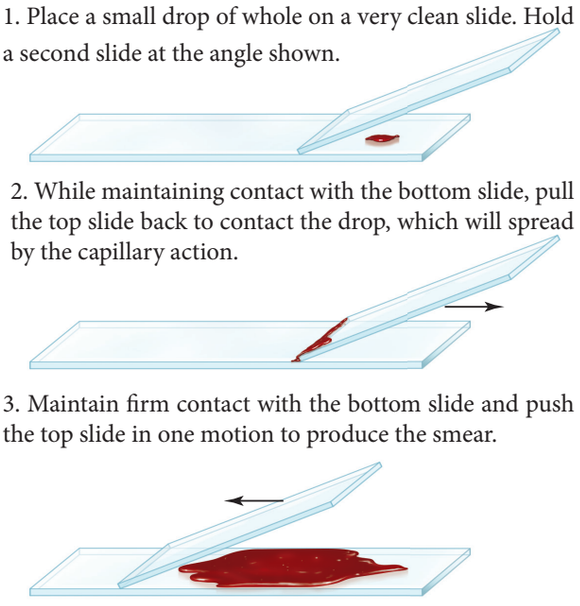


Figure 12.11 Peripheral blood smear preparation

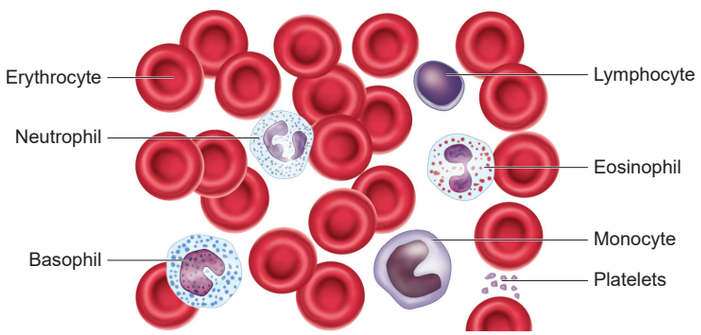


Figure 12.12 Smear of peripheral blood

Cell	Differential Count
Neutrophil	60-65%
Eosinophil	2-3%
Basophil	0.5-1.0%
Monocyte	1-3%
Lymphocyte	20-30%

Figure 12.13 Differential Count of White blood cells

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